

Installation and Setup

Installing PostgreSQL, pgAdmin, Configuration

1. Installing PostgreSQL

PostgreSQL can be installed on Windows, Linux, and macOS.

Steps (Windows):

1. Download PostgreSQL installer from the official website.
 2. Run the installer and select components:
 - PostgreSQL Server
 - pgAdmin
 - Command Line Tools
 3. Choose installation directory.
 4. Set **password for PostgreSQL superuser (postgres)**.
 5. Select port number (default: **5432**).
 6. Complete installation.
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2. Installing pgAdmin

pgAdmin is a graphical user interface (GUI) tool used to manage PostgreSQL databases.

- It is usually installed automatically with PostgreSQL.
 - If not installed, it can be downloaded separately.
 - pgAdmin allows:
 - Creating databases
 - Running SQL queries
 - Managing users and permissions
 - Viewing tables, schemas, and logs
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3. PostgreSQL Configuration

After installation, basic configuration is required.

a) Server Connection

- Open **pgAdmin**
- Connect to the PostgreSQL server using:
 - Host: localhost

- Port: 5432
- Username: postgres
- Password: (set during installation)

b) Configuration Files

- **postgresql.conf** – controls settings like memory, port, logging
- **pg_hba.conf** – controls client authentication and access rules

c) Creating a Database

CREATE DATABASE mydatabase;

d) Checking Server Status

- Ensure PostgreSQL service is running
- Can be checked via:
 - Services (Windows)
 - pg_ctl status